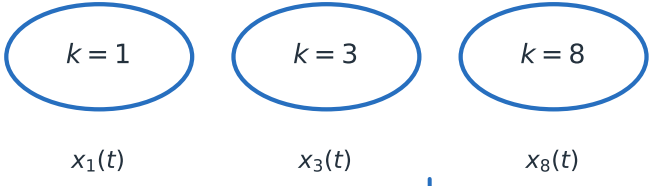
Degree-level and node-level network models

Both can be heterogeneous; they differ in state resolution and the network operator applied to infection pressure.

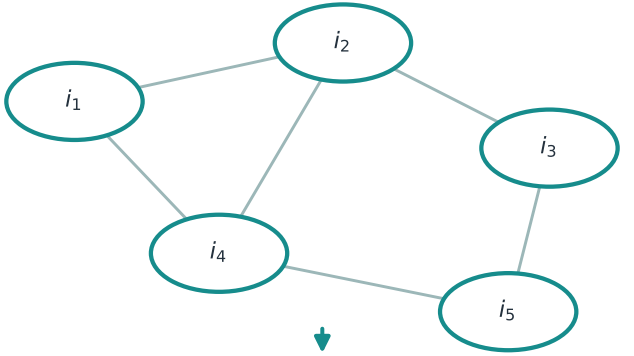
Degree-level state $x_k(t)$



mixing matrix $M_{k\ell}$
weights degree-class exposure

Cost scales with number of degree classes and time points

Node-level state $x_i(t)$



adjacency A_{ij}
preserves explicit contacts

Cost scales with nodes, edges, controls and time points